

NOTACIÓN SIGMA Y SUMAS DE RIEMMAN

Propósitos: Utilizar la notación sigma para escribir y calcular una suma /
Aproximar el área bajo la curva de la función $f(x)$ en el intervalo $[a, b]$.

NOTACIÓN SIGMA

Definición: Si $a_m, a_{m+1}, \dots, a_n$ son números reales y m y n son enteros tales que $m \leq n$, entonces

$$\sum_{i=m}^n a_i = a_m + a_{m+1} + a_{m+2} + \dots + a_{n-2} + a_{n-1} + a_n$$

$$\sum_{i=m}^n a_i$$

(La suma de a_i de $i = m$ hasta $i = n$)

n : Límite superior de la sumatoria.

m : Límite inferior de la sumatoria.

i : Índice de la sumatoria.

El símbolo $\sum_{i=m}^n$ indica una suma en la que la letra i , índice de la sumatoria o variable de la sumatoria, toma valores consecutivos que inician con m y terminan con n , es decir, $m, m + 1, m + 2, \dots, n - 2, n - 1, n$.

Ejemplo: Determinar

$$\sum_{j=0}^6 2^j$$

$$\sum_{j=0}^6 2^j = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6$$

$$\sum_{j=0}^6 2^j = 1 + 2 + 4 + 8 + 16 + 32 + 64$$

$$\sum_{j=0}^6 2^j = 127$$

$$\sum_{i=1}^5 2$$

$$\sum_{i=1}^5 2 = 2 + 2 + 2 + 2 + 2$$

$$\sum_{i=1}^5 2 = 10$$

Ejemplo: Escribir en notación sigma la siguiente suma

$$2^3 + 3^3 + 4^3 + \cdots + 11^3 + 12^3$$

$$\sum_{i=2}^{12} i^3$$

$$\sum_{i=1}^{11} (i + 1)^3$$

$$\sum_{i=0}^{10} (i + 2)^3$$

$$\sum_{i=-1}^9 (i + 3)^3$$

PROPIEDADES DE LA SUMA EMPLEANDO LA NOTACIÓN SIGMA

Teorema: Si c es cualquier constante, es decir, no depende de i , entonces

$$\sum_{i=m}^n c a_i = c \sum_{i=m}^n a_i$$

$$\sum_{i=m}^n (a_i - b_i) = \sum_{i=m}^n a_i - \sum_{i=m}^n b_i$$

$$\sum_{i=m}^n (a_i + b_i) = \sum_{i=m}^n a_i + \sum_{i=m}^n b_i$$

FÓRMULAS DE LA SUMA EMPLEANDO LA NOTACIÓN SIGMA

Teorema: Sea c una constante y n un entero positivo. Entonces

$$\sum_{i=1}^n 1 = 1n$$

$$\sum_{i=1}^n c = cn$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$$

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4} = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}$$

$$\sum_{i=1}^n i = 1 + \textcolor{brown}{2} + \textcolor{blue}{3} + \cdots + (\textcolor{brown}{n} - \textcolor{blue}{2}) + (\textcolor{blue}{n} - \textcolor{brown}{1}) + \textcolor{brown}{n}$$

$$\sum_{i=1}^n i = \textcolor{brown}{n} + (\textcolor{blue}{n} - \textcolor{brown}{1}) + (\textcolor{brown}{n} - \textcolor{blue}{2}) + \cdots + \textcolor{blue}{3} + \textcolor{brown}{2} + 1$$

$$2 \sum_{i=1}^n i = \underbrace{(\textcolor{brown}{n} + 1) + (\textcolor{brown}{n} + 1) + (\textcolor{brown}{n} + 1) + \cdots + (\textcolor{brown}{n} + 1) + (\textcolor{brown}{n} + 1) + (\textcolor{brown}{n} + 1)}_{\textcolor{brown}{n} \text{ términos}}$$

$$2 \sum_{i=1}^n i = \textcolor{brown}{n}(n+1)$$

$$\sum_{i=1}^n i = \frac{\textcolor{brown}{n}(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

Demostración por inducción matemática

1. Probar que para $n = 1$ es verdadera.

$$\sum_{i=1}^1 i^2 = 1^2 = 1$$

$$\frac{1(1+1)(2(1)+1)}{6} = \frac{1(2)(3)}{6} = 1$$

$$\sum_{i=1}^1 i^2 = 1^2 = 1 = \frac{1(1+1)(2(1)+1)}{6}$$

2. Probar que para $n = k$ es verdadera.

$$\sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$$

3. Probar que para $n = k + 1$ es verdadera.

$$\begin{aligned}\sum_{i=1}^{k+1} i^2 &= \sum_{i=1}^k i^2 + (k+1)^2 \\&= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\&= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\&= \frac{(k+1)}{6} [k(2k+1) + 6(k+1)] \\&= \frac{(k+1)}{6} [2k^2 + k + 6k + 6] \\&= \frac{(k+1)}{6} [(2k+3)(k+2)] \\&= \frac{(k+1)}{6} [(2k+2+1)(k+2)] \\&= \frac{(k+1)[(2(k+1)+1)(k+1+1)]}{6}\end{aligned}$$

Ejercicio: Evaluar

$$\begin{aligned}& \sum_{i=1}^n i(4i^2 - 3) \\&= \sum_{i=1}^n (4i^3 - 3i) \\&= \sum_{i=1}^n 4i^3 - \sum_{i=1}^n 3i \\&= 4 \sum_{i=1}^n i^3 - 3 \sum_{i=1}^n i \\&= 4 \left[\frac{n(n+1)}{2} \right]^2 - 3 \left[\frac{n(n+1)}{2} \right] \\&= n^2(n+1)^2 - \frac{3n(n+1)}{2} \\&= \frac{2n^2(n+1)^2 - 3n(n+1)}{2} \\&= \frac{n(n+1)[2n(n+1) - 3]}{2} \\&= \frac{n(n+1)[2n^2 + 2n - 3]}{2}\end{aligned}$$

Ejercicio: Evaluar

$$\begin{aligned}& \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \left[\left(\frac{i}{n} \right)^2 + 1 \right] \\&= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \left[\frac{i^2}{n^2} + 1 \right] \\&= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{3i^2}{n^3} + \frac{3}{n} \right]\end{aligned}$$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \frac{3i^2}{n^3} + \sum_{i=1}^n \frac{3}{n} \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{3}{n^3} \sum_{i=1}^n i^2 + \frac{3}{n} \sum_{i=1}^n 1 \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{3}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) + \frac{3}{n} n \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{3}{6} \left(\frac{n}{n} \times \frac{(n+1)}{n} \times \frac{(2n+1)}{n} \right) + 3 \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{1}{2} \left(\frac{n}{n} \times \frac{(n+1)}{n} \times \frac{(2n+1)}{n} \right) + 3 \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{1}{2} \left(1 \left(\frac{n}{n} + \frac{1}{n} \right) \left(\frac{2n}{n} + \frac{1}{n} \right) \right) + 3 \right] \\
&= \frac{1}{2} (1(1+0)(2+0)) + 3 \\
&= \frac{1}{2} (1(1)(2)) + 3 \\
&= 1 + 3 \\
&= 4
\end{aligned}$$

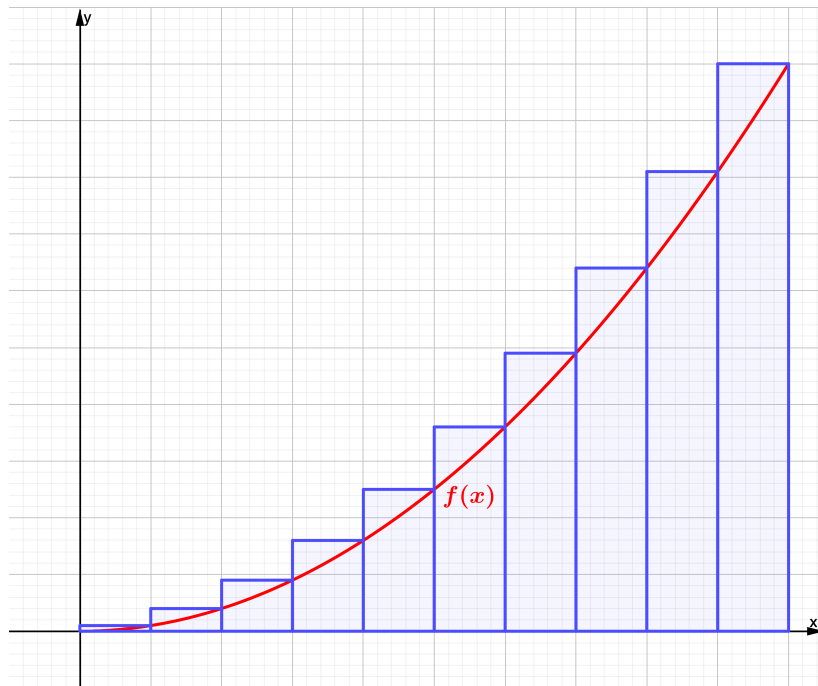
SUMAS DE RIEMMAN¹

Recuerde que al definir la pendiente de una recta tangente, primero obtuvimos aproximaciones de las pendientes de rectas secantes y, a continuación, tomamos el **límite** de estas aproximaciones. Sigamos una idea similar para las áreas. En primer lugar obtenemos una aproximación de la región S representándola por medio de rectángulos, y después tomamos el límite de las áreas de los rectángulos cuando se incrementa el número de éstos.

Definición: El área A de la región S que se encuentra bajo la gráfica de la función continua f es el límite de la suma de las áreas de los rectángulos de aproximación:

$$A = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} [f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_{n-1})\Delta x + f(x_n)\Delta x]$$

¹ <https://www.geogebra.org/classic/dgbe5cfs>



$$\sum_{i=1}^n f(x_i)\Delta x = f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_{n-1})\Delta x + f(x_n)\Delta x$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i)\Delta x$$

$$\Delta x = \frac{b - a}{n}$$

b : Límite superior del intervalo $x_n = b$

a : Límite inferior del intervalo $x_0 = a$

n : Cantidad de subintervalos formados en el intervalo $[a, b]$

$f(x_i)$: Alturas para cada uno de los rectángulos

Δx : Longitud de la base de los rectángulos

$$x_i = a + i\Delta x$$

$$x_i = a + i\left(\frac{b - a}{n}\right) \text{ Puntos terminales derechos}$$

$$x_i = a + (i - 1)\left(\frac{b - a}{n}\right) \text{ Puntos terminales izquierdos}$$

Ejemplo: Determinar el área bajo la curva de la función $f(x) = 2x^2 + 1$ en el intervalo $[-1,3]$ (Puntos terminales derechos).

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

$$[-1,3] \quad \Delta x = \frac{b-a}{n} \quad \Delta x = \frac{3-(-1)}{n} = \frac{4}{n}$$

$$x_i = a + i \left(\frac{b-a}{n} \right) \quad x_i = -1 + \frac{4}{n} i$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[f \left(-1 + \frac{4i}{n} \right) \right] \left(\frac{4}{n} \right)$$

$$f(x) = 2x^2 + 1, \quad f \left(-1 + \frac{4i}{n} \right) = 2 \left(-1 + \frac{4i}{n} \right)^2 + 1$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[2 \left(-1 + \frac{4i}{n} \right)^2 + 1 \right] \left(\frac{4}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[2 \left(1 - \frac{8i}{n} + \frac{16i^2}{n^2} \right) + 1 \right] \left(\frac{4}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[2 - \frac{16i}{n} + \frac{32i^2}{n^2} + 1 \right] \left(\frac{4}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[3 - \frac{16i}{n} + \frac{32i^2}{n^2} \right] \left(\frac{4}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{12}{n} - \frac{64i}{n^2} + \frac{128i^2}{n^3} \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \frac{12}{n} - \sum_{i=1}^n \frac{64i}{n^2} + \sum_{i=1}^n \frac{128i^2}{n^3} \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{12}{n} \sum_{i=1}^n 1 - \frac{64}{n^2} \sum_{i=1}^n i + \frac{128}{n^3} \sum_{i=1}^n i^2 \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{12}{n} (n) - \frac{64}{n^2} \left(\frac{n(n+1)}{2} \right) + \frac{128}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[12 - \frac{64}{2} \left(\frac{n}{n} \times \frac{(n+1)}{n} \right) + \frac{128}{6} \left(\frac{n}{n} \times \frac{(n+1)}{n} \times \frac{(2n+1)}{n} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[12 - 32 \left(1 \times \left(\frac{n}{n} + \frac{1}{n} \right) \right) + \frac{64}{3} \left(1 \times \left(\frac{n}{n} + \frac{1}{n} \right) \times \left(\frac{2n}{n} + \frac{1}{n} \right) \right) \right]$$

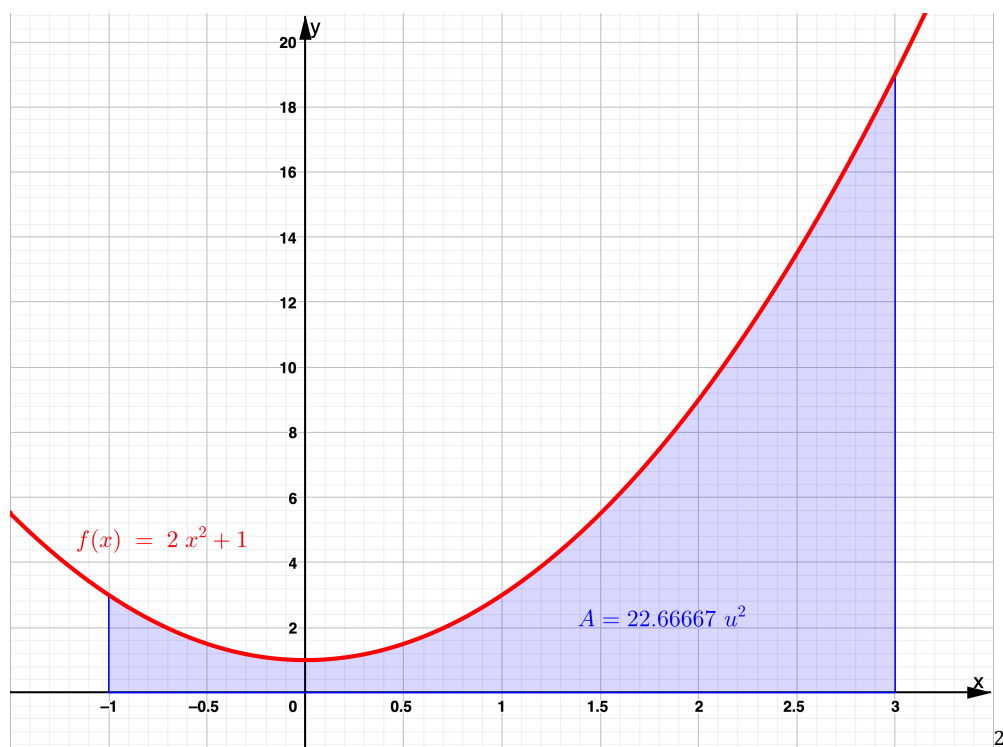
$$A = 12 - 32(1 \times (1 + 0)) + \frac{64}{3}(1 \times (1 + 0) \times (2 + 0))$$

$$A = 12 - 32(1) + \frac{64}{3}(1(1)(2))$$

$$A = 12 - 32 + \frac{128}{3}$$

$$A = \frac{36 - 96 + 128}{3}$$

$$A = \frac{68}{3} \approx 22, \bar{6}$$



Ejemplo: Determinar el área bajo la curva de la función $f(x) = 2x^2 + 1$ en el intervalo $[-1,3]$ (Puntos terminales izquierdos).

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

$$[-1,3] \quad \Delta x = \frac{b-a}{n} \quad \Delta x = \frac{3 - (-1)}{n} = \frac{4}{n}$$

$$x_i = a + (i-1) \left(\frac{b-a}{n} \right) \quad x_i = -1 + \frac{4i-4}{n}$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[f \left(-1 + \frac{4i-4}{n} \right) \right] \left(\frac{4}{n} \right)$$

$$f(x) = 2x^2 + 1, \quad f \left(-1 + \frac{4i-4}{n} \right) = 2 \left(-1 + \frac{4i-4}{n} \right)^2 + 1$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[2 \left(-1 + \frac{4i-4}{n} \right)^2 + 1 \right] \left(\frac{4}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[2 \left(1 - \frac{8i-8}{n} + \frac{16i^2 - 32i + 16}{n^2} \right) + 1 \right] \left(\frac{4}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[2 - \frac{16i-16}{n} + \frac{32i^2 - 64i + 32}{n^2} + 1 \right] \left(\frac{4}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[3 - \frac{16i-16}{n} + \frac{32i^2 - 64i + 32}{n^2} \right] \left(\frac{4}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{12}{n} - \frac{64i-64}{n^2} + \frac{128i^2 - 256i + 128}{n^3} \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \frac{12}{n} - \sum_{i=1}^n \frac{64i}{n^2} + \sum_{i=1}^n \frac{64}{n^2} + \sum_{i=1}^n \frac{128i^2}{n^3} - \sum_{i=1}^n \frac{256i}{n^3} + \sum_{i=1}^n \frac{128}{n^3} \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{12}{n} \sum_{i=1}^n 1 - \frac{64}{n^2} \sum_{i=1}^n i + \frac{64}{n^2} \sum_{i=1}^n 1 + \frac{128}{n^3} \sum_{i=1}^n i^2 - \frac{256}{n^3} \sum_{i=1}^n i + \frac{128}{n^3} \sum_{i=1}^n 1 \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{12}{n} \sum_{i=1}^n 1 - \frac{64}{n^2} \sum_{i=1}^n i + \frac{64}{n^2} \sum_{i=1}^n 1 + \frac{128}{n^3} \sum_{i=1}^n i^2 - \frac{256}{n^3} \sum_{i=1}^n i + \frac{128}{n^3} \sum_{i=1}^n 1 \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{12}{\textcolor{red}{n}}(\textcolor{red}{n}) - \frac{64}{n^2} \left(\frac{n(n+1)}{2} \right) + \frac{64}{n^{\textcolor{red}{2}}}(\textcolor{red}{n}) + \frac{128}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) - \frac{256}{n^3} \left(\frac{n(n+1)}{2} \right) + \frac{128}{n^{\textcolor{red}{3}}}(\textcolor{red}{n}) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[12 - \frac{64}{2} \left(\frac{n}{n} \times \frac{(n+1)}{n} \right) + \frac{64}{n} + \frac{128}{6} \left(\frac{n}{n} \times \frac{(n+1)}{n} \times \left(\frac{2n+1}{n} \right) \right) - \frac{256}{n} \left(\frac{n}{n} \times \frac{(n+1)}{n} \right) + \frac{128}{n^2} \right]$$

$$A = \lim_{n \rightarrow \infty} \left[12 - \frac{64}{2} \left(\frac{n}{n} \times \left(\frac{n}{n} + \frac{\textcolor{violet}{1}}{\textcolor{violet}{n}} \right) \right) + \frac{64}{n} + \frac{128}{6} \left(1 \times \left(\frac{n}{n} + \frac{\textcolor{violet}{1}}{\textcolor{violet}{n}} \right) \times \left(\frac{2n}{n} + \frac{\textcolor{violet}{1}}{\textcolor{violet}{n}} \right) \right) - \frac{256}{n} \left(\frac{n}{n} \times \left(\frac{n}{n} + \frac{\textcolor{violet}{1}}{\textcolor{violet}{n}} \right) \right) + \frac{128}{n^2} \right]$$

$$A = 12 - \frac{64}{2} (1 \times (1 + 0)) + \textcolor{red}{0} + \frac{128}{6} (1 \times (1 + \textcolor{violet}{0}) \times (2 + \textcolor{violet}{0})) - \textcolor{red}{0} (1 \times (1 + \textcolor{violet}{0})) + \textcolor{red}{0}$$

$$A = 12 - \frac{64}{2} (1) + \frac{128}{6} (1(1)(2))$$

$$A = 12 - 32 + \frac{128}{3}$$

$$A = \frac{36 - 96 + 128}{3}$$

$$A = \frac{68}{3} \approx 22.\bar{6}$$

Ejemplo: Determinar el área bajo la curva de la función $f(x) = x^3 - 6x$ en el intervalo $[0,5]$ (Puntos terminales derechos).

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

$$[0,5] \quad \Delta x = \frac{b-a}{n} \quad \Delta x = \frac{5-0}{n} = \frac{5}{n}$$

$$x_i = a + i \left(\frac{b-a}{n} \right) \quad x_i = 0 + \frac{5}{n} i$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[f\left(\frac{5i}{n}\right) \right] \left(\frac{5}{n}\right)$$

$$f(x) = x^3 - 6x, \quad f\left(\frac{5i}{n}\right) = \left(\frac{5i}{n}\right)^3 - 6\left(\frac{5i}{n}\right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\left(\frac{5i}{n}\right)^3 - 6\left(\frac{5i}{n}\right) \right] \left(\frac{5}{n}\right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{125i^3}{n^3} - \frac{30i}{n} \right] \left(\frac{5}{n}\right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{625i^3}{n^4} - \frac{150i}{n^2} \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n \frac{625i^3}{n^4} - \sum_{i=1}^n \frac{150i}{n^2} \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{625}{n^4} \sum_{i=1}^n i^3 - \frac{150}{n^2} \sum_{i=1}^n i \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{625}{n^4} \left(\frac{n^2(n+1)^2}{4} \right) - \frac{150}{n^2} \left(\frac{n(n+1)}{2} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{625}{4} \left(\frac{n^2}{n^2} \times \frac{(n+1)^2}{n^2} \right) - \frac{150}{2} \left(\frac{n}{n} \times \frac{(n+1)}{n} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{625}{4} \left(1 \times \frac{n^2 + 2n + 1}{n^2} \right) - \frac{150}{2} \left(1 \times \left(\frac{n}{n} + \frac{1}{n} \right) \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{625}{4} \left(1 \times \left(\frac{n^2}{n^2} + \frac{2n}{n^2} + \frac{1}{n^2} \right) \right) - \frac{150}{2} \left(1 \times \left(1 + \frac{1}{n} \right) \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{625}{4} \left(1 \times \left(1 + \frac{2}{n} + \frac{1}{n^2} \right) \right) - \frac{150}{2} \left(1 \times \left(1 + \frac{1}{n} \right) \right) \right]$$

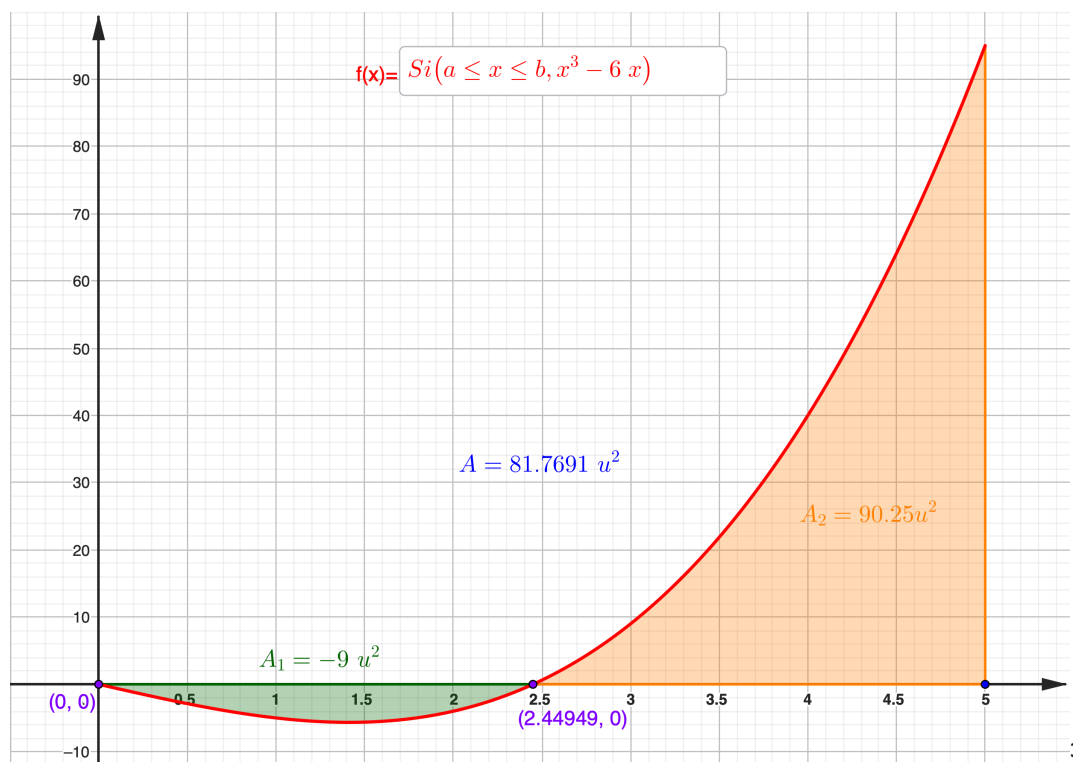
$$A = \frac{625}{4} (1 \times (1 + 0 + 0)) - \frac{150}{2} (1 \times (1 + 0))$$

$$A = \frac{625}{4} (1 \times (1)) - \frac{150}{2} (1 \times (1))$$

$$A = \frac{625}{4} - \frac{150}{2}$$

$$A = \frac{625}{4} - \frac{300}{4}$$

$$A = \frac{325}{4} \approx 81,25$$



Ejemplo: Determinar el área bajo la curva de la función $g(y) = 4y^2 - y^3$, $1 \leq y \leq 3$.

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n g(y_i) \Delta y$$

$$[1,3] \quad \Delta y = \frac{b-a}{n} \quad \Delta y = \frac{3-1}{n} = \frac{2}{n}$$

$$y_i = a + i \left(\frac{b-a}{n} \right) \quad y_i = 1 + \frac{2}{n} i$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[g \left(1 + \frac{2i}{n} \right) \right] \left(\frac{2}{n} \right)$$

$$g(y) = 4y^2 - y^3, \quad g \left(1 + \frac{2i}{n} \right) = 4 \left(1 + \frac{2i}{n} \right)^2 - \left(1 + \frac{2i}{n} \right)^3$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[4 \left(1 + \frac{2i}{n} \right)^2 - \left(1 + \frac{2i}{n} \right)^3 \right] \left(\frac{2}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[4 \left(1 + \frac{4i}{n} + \frac{4i^2}{n^2} \right) - \left(1 + \frac{6i}{n} + \frac{12i^2}{n^2} + \frac{8i^3}{n^3} \right) \right] \left(\frac{2}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[4 + \frac{16i}{n} + \frac{16i^2}{n^2} - 1 - \frac{6i}{n} - \frac{12i^2}{n^2} - \frac{8i^3}{n^3} \right] \left(\frac{2}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[3 + \frac{10i}{n} + \frac{4i^2}{n^2} - \frac{8i^3}{n^3} \right] \left(\frac{2}{n} \right)$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{6}{n} + \frac{20i}{n^2} + \frac{8i^2}{n^3} - \frac{16i^3}{n^4} \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{6}{n} \sum_{i=1}^n 1 + \frac{20}{n^2} \sum_{i=1}^n i + \frac{8}{n^3} \sum_{i=1}^n i^2 - \frac{16}{n^4} \sum_{i=1}^n i^3 \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{6}{n} (n) + \frac{20}{n^2} \left(\frac{n(n+1)}{2} \right) + \frac{8}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) - \frac{16}{n^4} \left(\frac{n^2(n+1)^2}{4} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{6}{n}(n) + \frac{20}{n^2} \left(\frac{n(n+1)}{2} \right) + \frac{8}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) - \frac{16}{n^4} \left(\frac{n^2(n+1)^2}{4} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{6}{n}(n) + \frac{20}{2} \left(\frac{n(n+1)}{n^2} \right) + \frac{8}{6} \left(\frac{n(n+1)(2n+1)}{n^3} \right) - \frac{16}{4} \left(\frac{n^2(n+1)^2}{n^4} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[\frac{6}{n}(n) + \frac{20}{2} \left(\frac{n}{n} \times \frac{(n+1)}{n} \right) + \frac{8}{6} \left(\frac{n}{n} \times \frac{(n+1)}{n} \times \left(\frac{2n+1}{n} \right) \right) - \frac{16}{4} \left(\frac{n^2}{n^2} \times \frac{(n+1)^2}{n^2} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[6 + 10 \left(1 \times \left(\frac{n}{n} + \frac{1}{n} \right) \right) + \frac{8}{6} \left(\frac{n}{n} \times \left(\frac{n}{n} + \frac{1}{n} \right) \times \left(\frac{2n}{n} + \frac{1}{n} \right) \right) - \frac{16}{4} \left(\frac{n^2}{n^2} \times \frac{n^2 + 2n + 1}{n^2} \right) \right]$$

$$A = \lim_{n \rightarrow \infty} \left[6 + 10 \left(1 \times \left(\frac{n}{n} + \frac{1}{n} \right) \right) + \frac{8}{6} \left(\frac{n}{n} \times \left(\frac{n}{n} + \frac{1}{n} \right) \times \left(\frac{2n}{n} + \frac{1}{n} \right) \right) - \frac{16}{4} \left(\frac{n^2}{n^2} \times \left(\frac{n^2}{n^2} + \frac{2n}{n^2} + \frac{1}{n^2} \right) \right) \right]$$

$$A = 6 + 10(1 \times (1 + 0)) + \frac{8}{6}(1 \times (1 + 0) \times (2 + 0)) - \frac{16}{4}(1 \times (1 + 0 + 0))$$

$$A = 6 + 10 + \frac{8}{6}(2) - \frac{16}{4}(1)$$

$$A = 16 + \frac{8}{3} - \frac{16}{4}$$

$$A = \frac{192}{12} + \frac{32}{12} - \frac{48}{12}$$

$$A = \frac{224}{12} - \frac{48}{12}$$

$$A = \frac{176}{12} u^3$$

$$A = \frac{44}{3} u^3 \approx 14, \bar{6} u^2$$

EJERCICIOS

a.	Calcular $\sum_{i=51}^{150} i^2$
b.	Calcular $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{i}{n^2}$
c.	Calcular $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{2(i-1)}{n} \right)^2 \left(\frac{2}{n} \right)$
d.	Determinar el área bajo la curva de la función $f(x) = 3x^2 + 4x$ en el intervalo $[0,2]$ (<u>Puntos terminales derechos</u>).
e.	Determinar el área bajo la curva de la función $f(x) = 6x^2 - 4$ en el intervalo $[2,5]$ (<u>Puntos terminales derechos</u>).
f.	Determinar el área bajo la curva de la función $f(x) = 2x + 1$ en el intervalo $[a, b]$ (a, b constantes con $a < b$) (<u>Puntos terminales derechos</u>).

RESULTADOS DE LOS EJERCICIOS

a.	1,093,350
b.	$\frac{1}{2}$
c.	$\frac{8}{3}$
d.	$A = 16$
e.	$A = 222$
f.	$A = b^2 + b - a^2 - a$

